

A Relational Observable Substrate and its NCG Realization

One Spectral Injection, the Connes–Chern Character, and the Disclinated Haldane Class

Working draft — two-stage formulation

local observable algebras and strict descent first; a single Dirac operator as the entire NCG
content;

the crystalline equivalence as foundational hypothesis

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Abstract

The relational origin is a *local observable net* $\mathcal{A} : \text{Reg} \rightarrow C^*\text{Alg}$, $O \mapsto \mathcal{A}(O)$, whose Stage-1 fibre over a patch U_i is a full local observable algebra $A_i = \mathcal{A}(U_i)$ — not a pre-selected occupied subspace. The occupied subspace, the Fermi projection, states, measurement effects and local unitaries are all *selected inside* A_i only after a realization or readout has been chosen. The relational layer is built as *strict descent data*: to each patch one attaches a local algebra-sheaf $\mathcal{A}_i$ on U_i , glued by comparison $*$ -isomorphisms on the overlaps, $\alpha_{ij} : \mathcal{A}_i|_{U_{ij}} \xrightarrow{\sim} \mathcal{A}_j|_{U_{ij}}$, satisfying the strict cocycle $\alpha_{ij}\alpha_{jk} = \alpha_{ik}$ on triple overlaps. The relational sheaf $F_{\text{rel}}(U_i) = \mathcal{A}_i$, glued by the α_{ij} , is genuinely relational — and flat — before it is spectral; the groupoid $R_Q \rightrightarrows I$ survives only as a presentation of the comparison site $K = N(R_Q)$, no longer foundational. Noncommutative geometry enters at exactly one place: a realization functor $\pi_{\text{NCG}} : R_Q \rightarrow \text{Hilb}_{A,D}$ that represents the local algebras and implements the comparisons,

$$A_i \mapsto (\pi_i : A_i \rightarrow B(\mathcal{H}_i)), \quad \alpha_{ij} \mapsto V_{ij}, \quad V_{ij} \pi_i(a) V_{ij}^* = \pi_j(\alpha_{ij}(a)),$$

and whose single essential datum is a Dirac operator D ; it realizes the sheaf as $U_i \mapsto P_i \Omega_D^\bullet(A_G) P_i$. The entire noncommutative content of the theory is this one relational Dirac operator. Because π_{NCG} is a functor the comparisons stay strictly composable, $V_{ij}V_{jk} = V_{ik}$, so the substrate carries no curvature; curvature is born in the realization, the first time the differential $d_D = [D, \cdot]$ acts. The connection is $a_{ij} = d_D V_{ij} = [D, V_{ij}]$ and the curvature is the Grassmann/Bellissard form $F = P(d_D P)^2$, simplicially $F_{ijk}^{\text{rel}} = P_i[D, V_{ij}][D, V_{jk}]$. Closure is the Bianchi identity, automatic from $P^2 = P$ and the Leibniz rule — and it is strict composability of the implementers that protects it; no twisted algebra and no associativity theorem are needed. A characteristic map — a cyclic cocycle τ , here the Connes–Chern/Bellissard area cocycle — sends the operator curvature to a scalar Čech–Connes cochain, with class $[c] \in \check{H}^2(N(R_Q); \mathbb{C})$ lifted to the integral $c_1^{\text{rel}} \in \check{H}^2(N(R_Q); \mathbb{Z})$; swapping τ targets higher Chern, $\mathbb{Z}_2$, gap-labeling, or Floquet invariants on the same substrate. Under a deferred crystalline comparison hypothesis c_1^{rel} recovers the Bloch–Berry Chern class of the periodic Haldane model. We localize c_1^{rel} to the star $\text{St}(d)$ of a pentagonal disclination and give a finite to-do list; notably the defect decision reduces to transport determinants and so does not require the full D . Two constructions are left honestly open: the relational Dirac operator (which carries all of the NCG content) and the realization-functor comparison.

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Architecture. Two stages, one injection:

$$\underbrace{\mathcal{A} : \text{Reg} \rightarrow C^* \text{Alg}, \quad F_{\text{rel}}(U_i) = \mathcal{A}_i, \quad \alpha_{ij}\alpha_{jk} = \alpha_{ik}}_{\text{Stage 1 + 1.5: local algebras, strict descent, flat}} \xrightarrow[\text{supply } D]{\pi_{\text{NCG}}} \underbrace{a_{ij} = [D, V_{ij}], \quad F = P(d_D P)^2, \quad \text{Bianchi free}}_{\text{Stage 2: NCG, curvature born at } d_D} \xrightarrow{\chi_r} c_1^{\text{rel}} \in \check{H}^2(N(R_Q); \mathbb{Z})$$

1 Stage 1: the local observable substrate

The substrate records what the local quantum world is made of, using observable algebras rather than a pre-selected sector. It is built in two strictly flat layers: a *local observable net* that assigns an algebra to each region (the carrier, §1.1), and a *strict relational descent layer* that glues these into a sheaf of observable algebras over a comparison site (Stage 1.5, §1.2). Both layers are curvature-free by design; the comparisons compose on the nose, so the substrate is flat before it is spectral.

1.1 The local observable net (the carrier)

Let Reg be the poset of admissible regions/patches, ordered by inclusion. The carrier is a local observable net — a functor

$$\mathcal{A} : \text{Reg} \longrightarrow C^* \text{Alg}, \quad O \longmapsto \mathcal{A}(O), \quad (1)$$

isotonic: an inclusion $O \subseteq O'$ induces a unital embedding $\mathcal{A}(O) \hookrightarrow \mathcal{A}(O')$, which already furnishes the restriction maps a sheaf needs. We deliberately do *not* impose the full Haag–Kastler AQFT

package (microcausality, covariance, a vacuum); only the algebraic carrier role is kept: local regions/patches carry observable algebras. The minimal load-bearing content of a patch U_i is its local observable algebra

$$A_i := \mathcal{A}(U_i), \quad (2)$$

a unital C^* -algebra. Occupied subspaces, Fermi projections, states, measurement effects and local unitaries are *not* primitive: they are selected inside A_i only after a realization/readout has been chosen (§2).

Remark 1.1 (Sector selection is downstream). The previous formulation took the primitive fibre to be an occupied subspace C_i (a projector P_i). That is now a Chern-sector reduction: choose a local Hamiltonian $H_i \in A_i$, form the Fermi projection $P_i = \chi_{(-\infty, E_F)}(H_i) \in \pi_i(A_i)''$ and the occupied subspace $E_i = \text{Ran } P_i$. The general relational carrier A_i is thereby separated from the specific Chern readout:

$$\text{local observable algebra} \rightsquigarrow \text{chosen projection} \rightsquigarrow \text{Chern/NCG curvature}.$$

1.2 Stage 1.5: the strict relational descent and the sheaf

On a relational cover $\{U_i\}$ the sheaf is built not from whole-algebra identifications but from genuine descent data. To each patch we attach a *local algebra-sheaf* $\mathcal{A}_i$ on U_i — the restriction of the net to subregions, $\mathcal{A}_i : V \mapsto \mathcal{A}(V)$ for $V \subseteq U_i$, which the isotony of §1.1 makes a presheaf (sheafified if needed). The relational structure is the gluing of these per-patch sheaves by comparison $*$ -isomorphisms defined *on the overlaps*,

$$\alpha_{ij} : \mathcal{A}_i|_{U_{ij}} \xrightarrow{\sim} \mathcal{A}_j|_{U_{ij}}, \quad (3)$$

satisfying the strict cocycle condition on triple overlaps,

$$\alpha_{ij} \alpha_{jk} = \alpha_{ik} \quad \text{on } U_{ijk}. \quad (4)$$

This is exactly descent data for a sheaf of observable algebras over the comparison site. When the cocycle holds strictly the data descend to a single global sheaf; the relational sheaf is the resulting local system of observable algebras,

$$F_{\text{rel}} : U_i \mapsto \mathcal{A}_i, \quad \text{glued by } \alpha_{ij}. \quad (5)$$

It is relational before it is spectral, and flat: the comparisons compose on the nose, so F_{rel} carries no holonomy. Whatever topology the system has is, at this stage, the combinatorial topology of the cover/site — coordinate-free, periodicity-free, and agnostic to which invariant one will later compute.

Remark 1.2 (The groupoid is a presentation, not a foundation). The comparison site is the nerve $K = N(R_Q)$ of the relational groupoid $R_Q \rightrightarrows I$; but R_Q now enters only as a convenient presentation of the cover and its overlaps, not as the primitive object. The descent data $(\mathcal{A}_i, \alpha_{ij})$ are foundational; the groupoid is bookkeeping. This replaces “groupoid of contexts” by “strict descent over a comparison site.”

Remark 1.3 (The defect is already in the descent site). At the $+\pi/3$ disclination the pentagon’s isotropy carries five composable comparisons, not six, so the cover around the defect — and hence $K = N(R_Q)$ — is structurally altered there before any operator is assigned. The relational topology sees the defect; the NCG realization will only measure it.

2 Stage 2: the NCG realization and the single injection D

Only after the relational sheaf exists does one choose a realization. This is the one and only place noncommutative geometry enters.

The functor. A realization is a functor $\pi_{\text{NCG}} : R_Q \rightarrow \text{Hilb}_{A,D}$ that represents the local algebras and implements the comparisons,

$$A_i \mapsto (\pi_i : A_i \rightarrow B(\mathcal{H}_i)), \quad \alpha_{ij} \mapsto V_{ij}, \quad R_Q \mapsto A_G, \quad (6)$$

with V_{ij} the (Kato) transport intertwining the comparison, $V_{ij}\pi_i(a)V_{ij}^* = \pi_j(\alpha_{ij}(a))$, and A_G the groupoid (Bellissard observable) algebra on $\mathcal{H} = \bigoplus_i \mathcal{H}_i$ with faithful trace T . The occupied sector is selected inside the represented algebra, $P_i \in \pi_i(A_i)''$, $E_i = \text{Ran } P_i$. The realized sheaf is the compressed local calculus $U_i \mapsto P_i \Omega_D^\bullet(A_G) P_i$; restriction is compression by sub-projectors (no surgery on the nonlocal D), gluing is by the transports.

The essential datum is D . Of everything π_{NCG} does, only one piece carries geometry: the Dirac operator D , equivalently the differential $d_D a = [D, a]$. The maps $A_i \mapsto \pi_i$ and $\alpha_{ij} \mapsto V_{ij}$ merely make the relations concrete; they inject no curvature. Hence the thesis of the paper: the entire noncommutative content of the theory is the single relational Dirac operator D ; supply it and the rest of the machinery turns automatically.

Remark 2.1 (Functoriality keeps the substrate flat). Because π_{NCG} preserves composition, $V_{ij}V_{jk} = V_{ik}$ holds strictly, so the triangular holonomy $V_{ki}V_{jk}V_{ij} = \mathbf{1}$ is trivial: a strict realization leaves no holonomy-curvature. All curvature must therefore come from D . This is the point, not a defect: it commits the construction to the spectral picture and, as §3 shows, it is exactly what protects the free Bianchi identity. On a defect, D cannot use global coordinates — see §8.

3 Connection, curvature, and the automatic Bianchi identity

Curvature is born here, when d_D first acts. The connection is the spectral derivative of the transport and the curvature is its compressed square:

$$a_{ij} := [D, V_{ij}] = d_D V_{ij}, \quad F = P(d_D P)^2 = P[D, P][D, P], \quad F_{ijk}^{\text{rel}} = P_i a_{ij} a_{jk} = P_i [D, V_{ij}][D, V_{jk}]. \quad (7)$$

Proposition 3.1 (Bianchi is automatic). *The curvature (7) is closed without any associativity hypothesis.*

(i) *From $P^2 = P$ one gets $P(d_D P)P = 0$, and with Leibniz this is the Bianchi identity $\nabla F = 0$ for the Grassmann connection $\nabla = P d_D$.*

(ii) *From the strictly preserved composition $V_{ij}V_{jk} = V_{ik}$ and Leibniz,*

$$[D, V_{ik}] = [D, V_{ij}] V_{jk} + V_{ij} [D, V_{jk}],$$

the Maurer–Cartan relation for a_{ij} , whence $\tilde{\delta} F_{ijk}^{\text{rel}} = 0$.

Proof. Differentiating $P^2 = P$ gives $(d_D P)P + P(d_D P) = d_D P$; compressing by P yields $P(d_D P)P = 0$, and the standard projector computation gives $\nabla F = 0$. Part (ii) is Leibniz applied to the functorial composition law of (6). $\square$

The earlier (holonomy) formulation had to earn closure through a twisted-algebra associativity theorem; here closure is a calculus identity, and Proposition 3.1(ii) makes explicit that it is functoriality of π_{NCG} — preservation of $V_{ij}V_{jk} = V_{ik}$ — that carries it. No second algebra is built.

4 The characteristic map and the relational Chern class

Closure of the operator curvature does not by itself yield a scalar; the bridge is a cyclic cocycle. This is the single, proper role of the characteristic map.

Definition 4.1 (Characteristic map). For a cyclic cocycle τ on A_G ($b\tau = 0$),

$$\chi_\tau : F_{ijk}^{\text{rel}} \mapsto c_{ijk} := \tau(F_{ijk}^{\text{rel}}) = \tau(P_i[D, V_{ij}][D, V_{jk}]) \in \mathbb{C}. \quad (8)$$

For the Chern/Hall family τ is the Connes–Chern (Bellissard area) cocycle, density $\frac{1}{2\pi i} \tau(P[\partial_1 P, \partial_2 P])$, $\partial_j = [X_j, \cdot]$.

Theorem 4.2 (Čech–Connes class). *If τ is a cyclic cocycle and F^{rel} is Bianchi-closed (Proposition 3.1), then $c = \chi_\tau(F^{\text{rel}})$ is a Čech 2-cocycle and defines $[c] \in \check{H}^2(N(R_Q); \mathbb{C})$ in the total cohomology of the Čech–Connes bicomplex (Appendix A). For the determinant/area cocycle the exponential sequence $U(1) \rightarrow \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{Z}$ integralizes the class to the relational first Chern class $c_1^{\text{rel}} \in \check{H}^2(N(R_Q); \mathbb{Z})$.*

The integral lift is the point: a Chern class is quantized, and that is visible only over $\mathbb{Z}$.

Remark 4.3 (The characteristic map is modular). The substrate, the realization, and Bianchi are fixed; the cyclic cocycle τ alone selects the invariant — area cocycle $\Rightarrow$ Chern/Hall, higher cocycles $\Rightarrow$ higher Chern, odd cocycles with a symmetry $\Rightarrow \mathbb{Z}_2$, the canonical trace $\Rightarrow$ gap-labeling, a quasi-energy $\tau \Rightarrow$ Floquet. The geometry never changes; only χ_τ does.

5 The crystalline equivalence (foundational hypothesis)

The validation against known topology — that in the translation-invariant sector the relational class is the Bloch–Berry Chern class — remains unproved and is recorded as the foundational hypothesis.

Hypothesis 5.1 (Crystalline equivalence). *For the periodic Haldane model there is a translation-invariant sector $\check{H}^2(N(R_Q); \mathbb{Z})_{\text{cryst}} \subseteq \check{H}^2(N(R_Q); \mathbb{Z})$ and a comparison $\Theta : \check{H}^2(N(R_Q); \mathbb{Z})_{\text{cryst}} \rightarrow H^2(T^2; \mathbb{Z})$ with $\Theta(c_1^{\text{rel}}) = c_1(B)$, $B \rightarrow T^2$ the Bloch bundle, carrying the Connes–Chern density to the Berry curvature.*

Corollary 5.2. *Assume Hypothesis 5.1. Then $\langle \Theta(c_1^{\text{rel}}), [T^2] \rangle = \langle c_1(B), [T^2] \rangle = C$, the Haldane Chern number.*

This is the external bridge to Bloch theory; it requires periodicity (a torus). The disclination class lies outside the crystalline sector and is reached without it.

6 The disclination and the relative class

Let d be a pentagonal disclination, $\text{St}(d) \subset N(R_Q)$ its star, $\partial \text{St}(d)$ its boundary. With c^{ref} the cochain of the defect-free reference, the relative cocycle $\rho_d = c - c^{\text{ref}}$ is supported on $\text{St}(d)$ when defect and reference agree outside it, and

$$c_d^{\text{rel}} := [\rho_d] \in \check{H}^2(\text{St}(d), \partial \text{St}(d)), \quad \Phi_d = \prod_{(ij) \in L_d} g_{ij}, \quad g_{ij} = \frac{\det V_{ij}}{|\det V_{ij}|}, \quad (9)$$

the pairing of c_d^{rel} with the loop L_d around the pentagon (Figure 1).

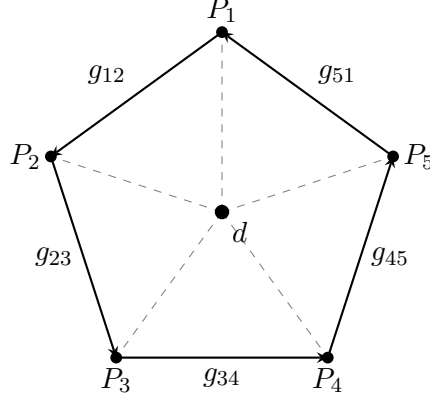


Figure 1: The star $\text{St}(d)$: five contexts, five transports, five triangles $\{d, i, i+1\}$ carrying $F_{d,i,i+1}^{\text{rel}}$. Curvature is supported here; c_1^{rel} localizes as c_d^{rel} , paired with L_d as $\Phi_d = \prod_i g_{i,i+1}$.

7 The Local Defect Theorem

Theorem 7.1 (Local Defect Theorem). *Assume bulk flatness: $F = F^{\text{ref}}$ and $P = P^{\text{ref}}$ off $\text{St}(d)$. Then c_d^{rel} is gauge- and Morita-invariant, supported on $\text{St}(d)$, and*

$$[c_d] \neq 0 \iff c_d^{\text{rel}} \neq 0 \iff \Phi_d \neq \Phi_d^{\text{ref}}.$$

Moreover the deciding quantity Φ_d is built from transport determinants g_{ij} alone, so for the determinant (Chern) characteristic map the defect decision is independent of the full D .

Sketch. Invariance: $[c]$ is a Čech–Connes class; a frame change shifts a_{ij} by an inner term traced into a coboundary by cyclicity of τ , and a realization change presenting the same R_Q is Morita. *Localization:* bulk flatness makes ρ_d vanish off $\text{St}(d)$; the determinant phases telescope along interior transports, leaving the boundary ratio $\Phi_d/\Phi_d^{\text{ref}}$. The last clause follows because $g_{ij} = \det V_{ij}/|\det V_{ij}|$ does not involve D . $\square$

So the defect monodromy is a transport-level quantity (post-functor, pre- d_D), while the bulk Chern class is a D -level quantity; this theorem is the bridge that lets the former decide the latter.

8 Two open constructions

The two-stage formulation concentrates all genuine difficulty in two places.

Open construction 8.1 (The relational Dirac operator — the whole NCG content). On a disclination there is no single-valued position, so D cannot be the flat-plane X, Y . It must be built from a position 1-cocycle on the groupoid, $X : R_Q \rightarrow \mathbb{R}^2$ with $X(\gamma\gamma') = X(\gamma) + \gamma \cdot X(\gamma')$, giving relational derivations $\partial_j = [X_j, \cdot]$ and hence D without global coordinates. By the thesis of §2 this single object carries the entire noncommutative content of the theory; its existence, properness, and the self-adjointness of D are what the construction pivots on.

Open construction 8.2 (The realization comparison). π_{NCG} induces a map on cohomology; one must show it is an isomorphism on the sector of interest, so that the relational Čech class and the realized NCG class agree. Equivalently, the bulk Bellissard curvature $P[D, P][D, P]$ and the simplicial F_{ijk}^{rel} represent the same Čech–Connes class. This comparison replaces the former associativity theorem as the one nontrivial internal identity; it is periodicity-free, and distinct from Hypothesis 5.1 (which is the external bridge to Bloch theory).

9 To-do list: computing the Haldane disclination

For the Haldane model in the topological regime ($C = \pm 1$) the pipeline is finite.

Stage 1 — local algebra substrate.

- Attach local observable algebras A_i around the defect; select $H_i \in A_i$, $P_i = \chi_{(-\infty, E_F)}(H_i)$, $E_i = \text{Ran } P_i$ (frames e_i if used). Record the comparison $*$ -isomorphisms α_{ij} on overlaps and the descent site $K = N(R_Q)$, including the five-fold star at d .

Stage 2 — realize and differentiate.

- Apply π_{NCG} : $\pi_i : A_i \rightarrow B(\mathcal{H}_i)$, $\alpha_{ij} \mapsto V_{ij}$ (Kato transport from sector overlaps), $R_Q \mapsto A_G$.
- Build the relational Dirac operator D from the position cocycle X (Open construction 8.1); form $a_{ij} = [D, V_{ij}]$ and $F_{ijk}^{\text{rel}} = P_i a_{ij} a_{jk}$.
- Closure is structural: verify $[D, V_{ik}] = [D, V_{ij}]V_{jk} + V_{ij}[D, V_{jk}]$ (Proposition 3.1); no associativity check.

Stage 3 — characteristic map and class.

- Apply the Connes–Chern area cocycle: $c_{ijk} = \tau(F_{ijk}^{\text{rel}})$; integralize to $c_1^{\text{rel}} = [c] \in \check{H}^2(N(R_Q); \mathbb{Z})$.
- Cross-check $\langle c_1^{\text{rel}}, [\Sigma] \rangle$ against the bulk Bellissard number $\frac{1}{2\pi i} \tau(P[\partial_1 P, \partial_2 P])$.

Stage 4 — localize and decide (transport-level).

- On $\text{St}(d)$ form $\rho_d = c - c^{\text{ref}}$ (nonzero by five- vs. six-fold coordination); $\text{St}(d)$ is a disk so $\check{H}^2(\text{St}(d), \partial \text{St}(d)) \cong \mathbb{Z}$.
- Compute the links $g_{i,i+1} = \det V_{i,i+1} / |\det V_{i,i+1}|$ and $\Phi_d = \prod_i g_{i,i+1}$, Φ_d^{ref} . The disclination conjugates the incident $V_{i,i+1}$ by $R(\pi/3)$ (the Frank wedge).

- Decide: $\Delta\Phi = \arg(\Phi_d/\Phi_d^{\text{ref}}) \in \mathbb{R}/2\pi\mathbb{Z}$, with $c_d^{\text{rel}} \neq 0 \iff \Delta\Phi \notin 2\pi\mathbb{Z}$. By Theorem 7.1 this step needs only the transports, not the full D ; the expected form is $\Delta\Phi \propto (\text{Frank angle}/2\pi)$ weighted by bulk band data.

10 Status and outlook

Relations come first and are flat; the realization functor injects geometry at one point, the Dirac operator D , and curvature is born only when d_D acts. Closure is then a calculus identity that functoriality protects (Proposition 3.1), and a single cyclic cocycle bridges operator curvature to a Čech–Connes class and selects the invariant. The crystalline equivalence (Hypothesis 5.1) remains the open external validation against Bloch theory and is not assumed. All genuine difficulty is concentrated in the relational Dirac operator (Open construction 8.1) and the realization comparison (Open construction 8.2); the disclination decision, by contrast, reduces to a finite transport-determinant holonomy once D is fixed.

A The Čech–Connes bicomplex

The cover direction supplies the Čech differential $\check{\delta}$ on $N(R_Q)$; the operator direction supplies the cyclic (b, B) differentials of $\Omega_D^\bullet(A_G)$. Curvature is a total representative: F^{rel} is $\check{\delta}$ -closed by Bianchi (Proposition 3.1) and τ is b -closed; χ_τ contracts the operator legs, leaving a $\check{\delta}$ -cocycle of class $[c] \in \check{H}^2(N(R_Q); \mathbb{C})$, integralized to c_1^{rel} . This is the standard setting for a relational NCG Chern–Weil statement: noncommutative curvature, abelian cohomology, mediated by a cyclic characteristic map [1, 2, 5].

B Dictionary: relational $\rightarrow$ realized $\rightarrow$ Haldane

Relational (Stage 1 / 1.5)	NCG realization (Stage 2)	Haldane
local algebra A_i	represented $\pi_i(A_i) \subset B(\mathcal{H}_i)$	local observable algebra
selected sector $P_i \in \pi_i(A_i)''$	$E_i = \text{Ran } P_i$	Fermi projection $\chi_{(-\infty, E_F)}(H_i)$
comparison α_{ij} (strict $*$ -iso)	implementer $V_{ij}, V_{ij}V_{jk} = V_{ik}$	Kato/Wilson link
descent site $K = N(R_Q)$	algebra A_G , Dirac D	observables, position cocycle
sheaf $F_{\text{rel}}(U_i) = \mathcal{A}_i$ (per-patch, glued by α_{ij})	$P_i \Omega_D^\bullet(A_G) P_i$	local calculi
(flat: strict descent)	connection $a_{ij} = [D, V_{ij}]$	transport derivative
—	curvature $F = P(d_D P)^2$	$\pi/3$ at the pentagon
combinatorial closure	$P^2 = P$ & Leibniz $\Rightarrow \check{\delta}F = 0$	automatic Bianchi
—	χ_τ (Connes–Chern)	area cocycle
relational class $[c] \in \check{H}^2$; lift $c_1^{\text{rel}} \in \check{H}^2(\mathbb{Z})$		class; defect c_d^{rel}

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